B12

Study of Optical Photons using Geant4 Simulation Toolkit

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IITR@175 Partnerships

Introduction

Scintillator detectors are an indispensable for energy determination of gamma rays. A high energy gamma beam is converted to optical photons in a scintillator. The optical photons fall on the PMT to generate photoelectrons. The photoelectrons acts as a proxy for the incident gamma. The energy of the gamma ray is determined as the pulse heights generated by the photoelectrons. Studying the generation, transport and detection of Optical Photons is significant for improving gamma detector efficiency. GEANT4 provides accurate radiation transport and optical photon transport simulations. It is therefore, ideal for simulation of scintillator. Initially, we generate OP's as primary particles, to better understand their physics. Simulations of scintillator detectors is performed to obtain pulse height spectra. The OP's are produced as secondaries, mainly due to Photoelectric effect and Compton Scattering.

Geant4

Geant4 (for GEometry ANd Tracking) is a simulation package developed by RD44, worldwide collaboration of institutes. It is widely used in high energy, nuclear and accelerator physics.

Fig 1: Geant4 Logo

User Input: Shape, Size, Material of detector, Particle Generator defined as separate classes.

Output: Accurate simulations of passage of particles through matter using Monte Carlo.

Offers great flexibility and ease for simulation and tracking of optical photons.

Optical Photons

Optical Photons: Having wavelength much greater than typical atomic spacing.

Production: Scintillation, Cherenkov Radiation

Processes: Refraction, Reflection, Bulk absorption, Mie and Rayleigh scattering.

Geant4: Optical Photons produced follows Gaussian distribution. The fluctuations about mean are controlled by RESOLUTIONSCALE parameter. Thus, **Intrinsic Resolution** has been incorporated.

Thus, it offers more insight and enables the investigation of the parameters which affect the physics processes of optical photons. Scintillation, optical, physical and surface properties can be optimized for new and more complicated settings, without resorting to experiments.

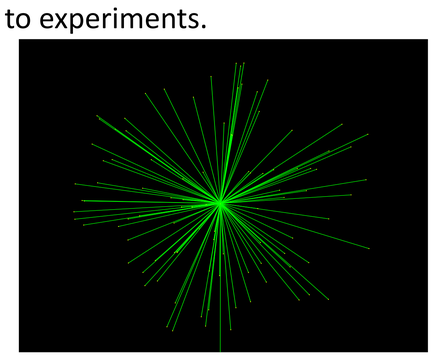


Fig 2: Generation of Optical Photons

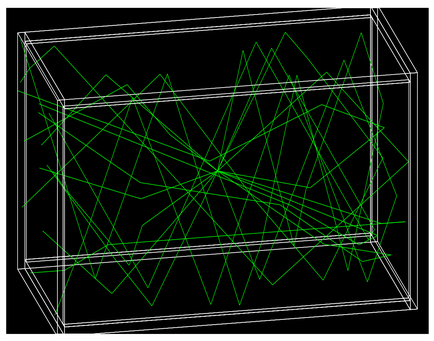
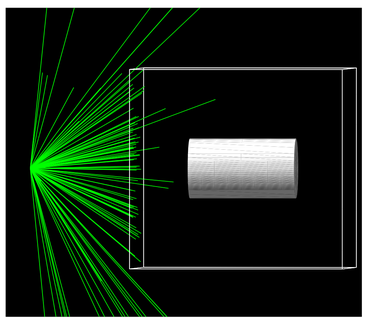
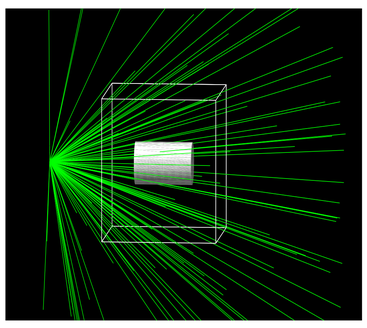


Fig 3: Reflection of OP's from Aluminum

Our Work

Shielding Effects





Detector Efficiency

Setup: 3.5" x 6" crystal. Isotropic gamma source kept at 1 mm from crystal.

PPE: 22.9 %

TE: 37.19 %

With 1 mm Al plate

PPE: 19.5 %

TE: 34.25 %

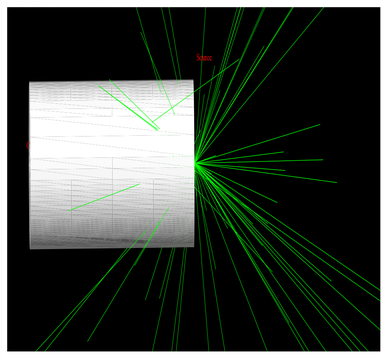
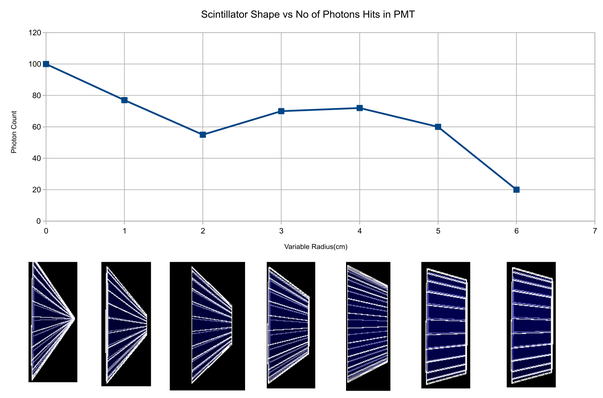


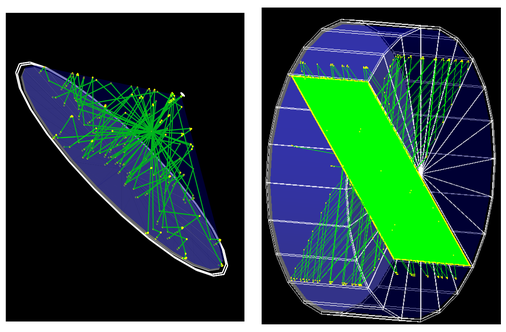
Fig 4: Setup for Efficiency Calculations

Scintillator Shape vs Photon count



Cone
ToA: 0.447 ns
OP : 96/100

Cylinder
ToA: 33 ms
OP : 59/100



The effect of scintillator shape on the light collection efficiency was studied by comparing Time of Arrival and Number of photons at the detector. Conical crystal had a better LCE than cylindrical

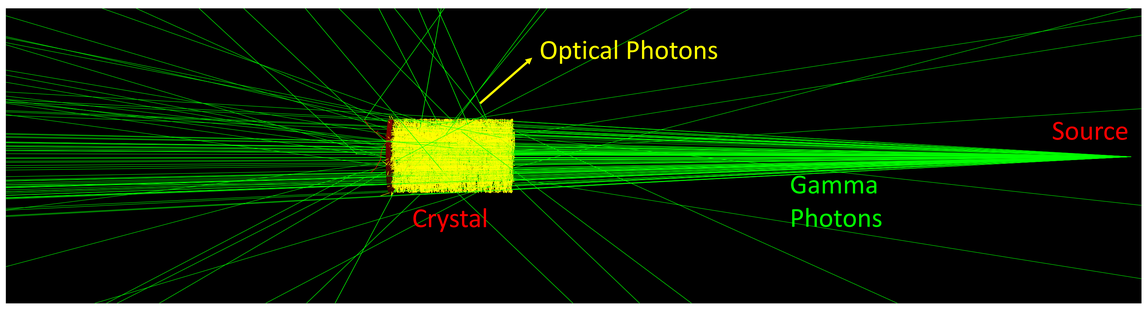
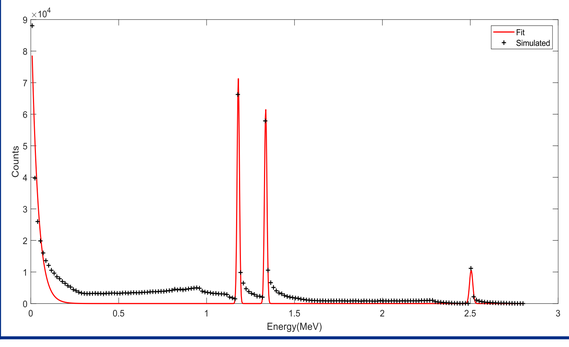


Fig 5: Simulation of Scintillation in Geant4

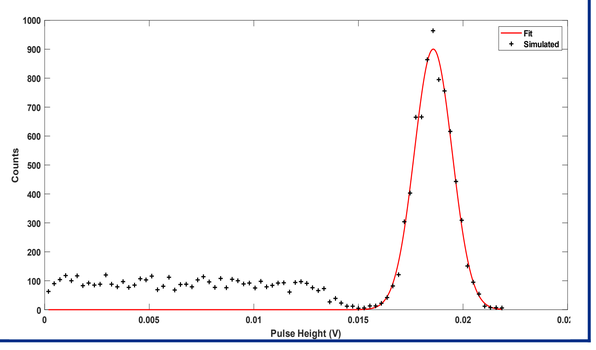
Pulse Height Spectra

A scintillator detector (2" x 2") was simulated. An isotropic gamma point source was modelled. The number of optical photons produced was obtained. Pulse Height Spectra were obtained from number of light photons. Energy spectra was also obtained directly from the energy deposited by each gamma photon.

Co60



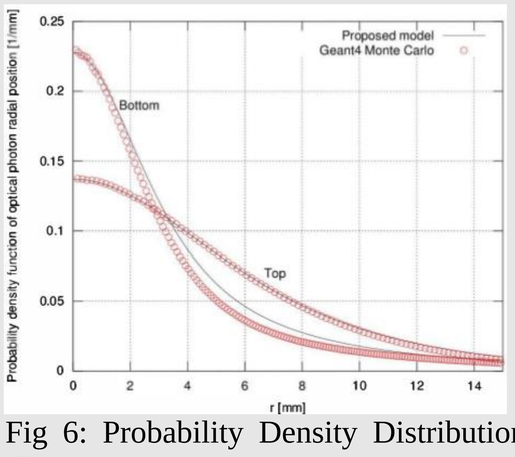
Cs 137



Literature Review

Geometric Model for radial distribution of photons on detector surface

A novel mathematical model for light propagation based on geometric optics laws.



Comparision for 1000 γ at 142 keV.

Proposed	Monte Carlo
< 1 sec	25 mins

Fig 6: Probability Density Distributions of photons

Effect of Scintillation Crystal shape on light collection efficiency

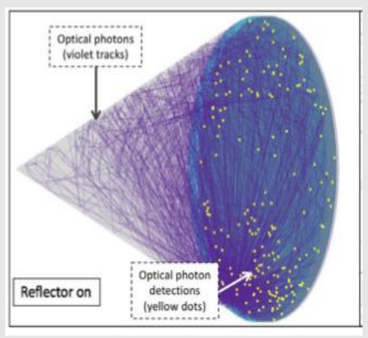


Fig 7: Optical Photons in a Conical Scintillators

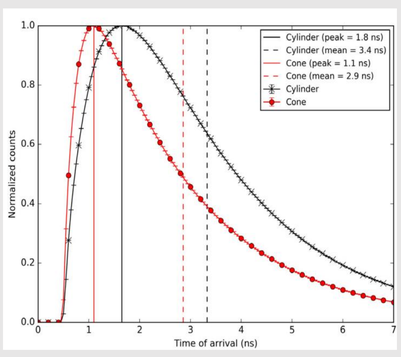
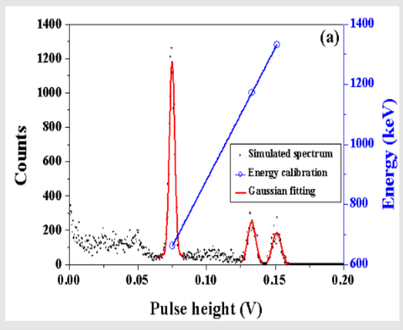


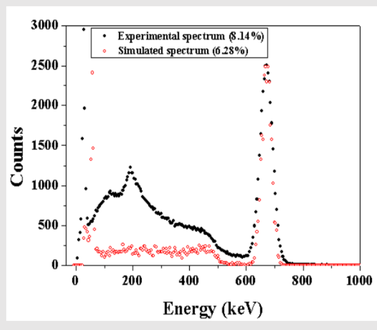
Fig 8: TOA distributions

Conical crystals have a higher LCE than cylindrical owing to reduced number of internal reflections in the conical crystal.

Simulated Pulse Height Spectra of Cs137 + Co60

A CsI(Tl), 2" x 2" scintillator was simulated. The simulated and the experimental spectra seems to agree well. The obtained resolution at 662 keV was: 6.28 %





Pulse height obtained via the number of photons produced in scintillation due to each gamma ray event.

Results and Future Work

The simulated pulse height spectra agrees well the literature. Various exercises with optical photons also obey physical laws. Detector efficiency for different configurations and different sources has been estimated through simulations. In future, we will work on obtaining sum peak efficiency, thereby getting three data points from each two-step cascade decay nuclides like Co60, Sc46, Nb94. Vidmar Efficiency calibration method will be verified theoretically/experimentally. Applications of optical photons in medical imaging will be explored further through simulations and experiments. A mathematical framework for incorporating sum peak efficiency in efficiency calibration curve of multi energetic sources.